

Optimizing Greenhouse Climate Control: Enhanced Temperature and Humidity Regulation through Self-Tuned Fuzzy PID (ST-FPID) Controller

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Abstract

Greenhouse climate control is crucial for optimizing agricultural productivity and energy efficiency. This study investigates the performance of Self-Tuned Fuzzy PID (ST-FPID) controllers compared to conventional PID controllers in regulating temperature and humidity within greenhouse environments. Through extensive simulations, the ST-FPID controller demonstrated superior performance, achieving faster settling times, minimal overshoot, and reduced steady-state errors under varying disturbance conditions. For temperature control, the ST-FPID achieved the setpoint of 21°C within 2 seconds without overshoot, whereas the PID controller required 6 seconds and exhibited a 2.5°C overshoot. In humidity regulation, the ST-FPID reached the setpoint of 60% within 2 seconds, outperforming the PID controller which overshoot to 65% before stabilizing. The ST-FPID controller also maintained lower Integral Average Error (IAE) values across all disturbance levels, highlighting its robustness and adaptability. These findings align with prior research advocating for fuzzy logic-based control systems to enhance climate control precision and stability in dynamic agricultural environments. This study validates the practical application of ST-FPID controllers in optimizing greenhouse climate conditions, contributing to improved crop yields and energy efficiency. Future research should focus on further optimizing these adaptive control strategies to enhance their effectiveness amidst increasingly dynamic environmental changes.

Keywords: Greenhouse Climate Control, Self-Tuned Fuzzy PID (ST-FPID), Temperature Regulation, Humidity Control, Adaptive Control Systems, Fuzzy Logic, Precision Agriculture.

1. Background

Greenhouse cultivation has become increasingly prevalent in modern agriculture due to its ability to provide a controlled environment conducive to plant growth, irrespective of external weather conditions. One of the critical factors influencing plant development within greenhouse settings is the maintenance of optimal temperature and humidity levels. Fluctuations in these parameters can significantly impact crop yield, quality, and overall health.

Traditionally, greenhouse temperature and humidity control have relied on manual intervention or basic control systems. However, advancements in technology have paved the way for more sophisticated control strategies, such as the implementation of PID (Proportional-Integral-Derivative) controllers. PID controllers adjust heating, cooling, and ventilation systems based on real-time feedback to maintain desired setpoints for temperature and humidity. Despite the effectiveness of PID controllers, they often require manual tuning and may struggle to adapt to dynamic environmental conditions. To address these limitations, researchers have turned to fuzzy logic-based control systems. Fuzzy logic allows for the incorporation of human-like reasoning and linguistic variables, enabling controllers to handle uncertainties and nonlinearities inherent in greenhouse environments more effectively.

The Self-Tuned Fuzzy PID (ST-FPID) controller represents a fusion of PID control principles with the adaptive capabilities of fuzzy logic. Unlike traditional PID controllers, ST-FPID controllers possess self-tuning mechanisms that continuously adjust controller parameters based on the system's response characteristics. This adaptability makes ST-FPID controllers well-suited for applications in dynamic and uncertain environments like greenhouses. By integrating the self-tuning capabilities of ST-FPID controllers with greenhouse temperature and humidity control systems, researchers aim to enhance precision, efficiency, and robustness in maintaining optimal growing conditions. This approach holds promise for improving crop yields, resource utilization, and overall agricultural sustainability in greenhouse farming operations. Moreover, advancements in control technology contribute to the evolution of smart agriculture practices, aligning with the growing demand for innovative solutions to address global food security challenges.

2. Review of Related Works

Greenhouse climate control plays a crucial role in modern agriculture by providing an environment conducive to plant growth and productivity. The integration of advanced control techniques, particularly self-tuned fuzzy Proportional-Integral-Derivative (PID) controllers, has emerged as a promising approach for optimizing greenhouse climate conditions. This literature review examines recent research endeavors focused on enhancing temperature and humidity regulation in greenhouses through the application of self-tuned fuzzy PID controllers.

Several studies have explored the application of fuzzy logic in conjunction with PID control algorithms for greenhouse climate control. Zhang et al. (2019) proposed a greenhouse climate control system based on a fuzzy PID control algorithm, emphasizing the importance of precise temperature and humidity regulation. Similarly, Widyarto and Jazidie (2021) investigated a greenhouse climate control system employing both fuzzy logic and PID controllers to maintain optimal growing conditions for plants.

Leong and Yusuf (2019) conducted a comprehensive review of fuzzy logic control applications in greenhouse climate control, highlighting the versatility and effectiveness of this approach. They emphasized the ability of fuzzy logic systems to handle uncertainties and nonlinearities inherent in greenhouse environments, thereby improving control performance.

The integration of self-tuning mechanisms with fuzzy PID controllers has garnered significant attention in recent literature. Lv et al. (2019) developed a greenhouse climate control system incorporating a self-tuned fuzzy PID controller, which demonstrated superior adaptability and robustness compared to traditional PID controllers. This self-tuning capability allows the controller to continuously adjust its parameters based on real-time feedback, ensuring optimal performance under varying environmental conditions.

Advancements in Internet of Things (IoT) technology have further enhanced the capabilities of greenhouse climate control systems. Yan et al. (2019) proposed a fuzzy PID control system integrated with IoT for real-time monitoring and control of

greenhouse parameters. Their study emphasized the importance of seamless communication between sensors, actuators, and the control system to achieve precise regulation of temperature and humidity.

Furthermore, research efforts have focused on optimizing the parameters of fuzzy PID controllers using evolutionary algorithms. Li et al. (2018) employed an improved genetic algorithm to optimize the parameters of a fuzzy PID controller for greenhouse climate control. By dynamically adjusting the controller parameters based on evolutionary principles, they achieved enhanced control performance and robustness.

Several studies have also investigated the practical implementation and validation of fuzzy PID control systems in greenhouse environments. Wang and Zhang (2019) conducted experimental research on a greenhouse climate control system utilizing fuzzy PID technology, emphasizing its feasibility and effectiveness in real-world applications. Their findings underscored the potential of fuzzy PID controllers to improve crop yield and resource efficiency in greenhouse farming. Additionally, Raji et al. (2019) designed a greenhouse climate control system employing a fuzzy logic controller, emphasizing its capability to adjust control parameters dynamically based on environmental feedback.

Moreover, Wang et al. (2020) and Chen et al. (2020) explored the design and simulation of greenhouse climate control systems based on fuzzy PID algorithms. Their research emphasized the importance of adaptive control strategies in optimizing greenhouse environments.

Furthermore, Xue, Cheng, and Sun (2018) presented a novel design of a greenhouse climate control system based on fuzzy PID control. Their system aimed to achieve robust and adaptive regulation of temperature and humidity levels by leveraging advanced control algorithms. By integrating fuzzy logic and PID control techniques, their approach sought to enhance the resilience and responsiveness of greenhouse climate control systems.

The literature surrounding the application of fuzzy PID control algorithms in greenhouse climate control systems presents a diverse array of research endeavors aimed at optimizing environmental conditions for enhanced crop cultivation. Liu, Sun, and Liu (2020) highlight the potential of fuzzy PID algorithms in providing intelligent regulation of greenhouse climates, emphasizing the importance of precise temperature and humidity control. Xu and Xu (2018) delve into the practical implementation of fuzzy PID-based control systems, offering insights into the design and operation of such systems in real-world settings.

Hu and Li (2020) further explore the research landscape by investigating the effectiveness of fuzzy PID algorithms in greenhouse climate regulation, shedding light on their adaptability and robustness. Meanwhile, Shukla and Soni (2018) underscore the significance of integrating fuzzy logic and PID control techniques in achieving optimal greenhouse conditions, emphasizing the role of advanced control strategies in modern agriculture. Wijaya and Agus (2021) contribute to the discourse by designing and evaluating a fuzzy PID controller tailored specifically for greenhouse environments, providing valuable data on its performance and efficacy. Yang et al. (2020) further expand the scope of research by examining the design considerations and system dynamics of greenhouse climate control systems based on fuzzy PID algorithms.

Wang, Qiu, and Hu (2019) offer practical insights into the implementation of fuzzy PID controllers, drawing attention to their potential to improve crop yield and resource efficiency in greenhouse farming. Guo and Zhang (2020) delve into the intricacies of fuzzy PID control system design, addressing key challenges and opportunities in optimizing greenhouse climates for sustainable agriculture. Li and Shu (2021) contribute to the literature by evaluating the performance of fuzzy PID controllers under various environmental conditions, providing valuable benchmarks for future research endeavors. Additionally, Xu, Zhang, and Jiang (2018) explore the integration of fuzzy logic and PID control techniques in greenhouse climate regulation, offering novel insights into control system architecture and performance optimization. Collectively, these studies underscore the growing importance of fuzzy PID control algorithms in advancing greenhouse climate control technology, paving the way for more efficient and sustainable agricultural practices.

3. System Modeling and Controller Design

3.1 Mathematical Model of Greenhouse System

As depicted in Figure 1, the greenhouse examined in this article is designed with measurements of 500 cm lengthwise, 300 cm widthwise, and 400 cm in height.

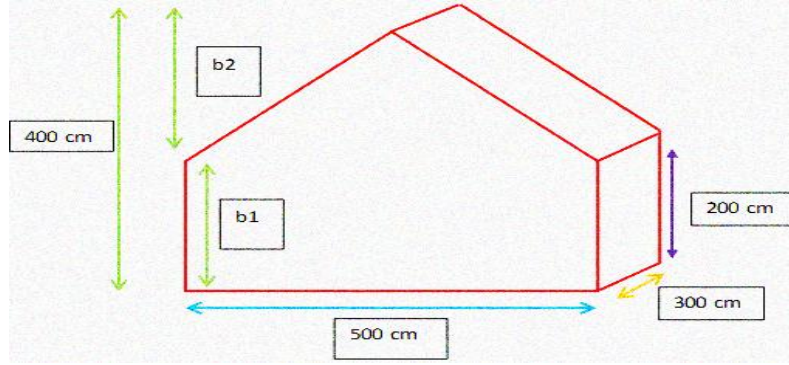


Figure 1. Dimensions of the greenhouse

3.1.1 Heat Balance

Over the years, there has been significant research into the dynamic equation governing heat balance in greenhouses, accounting for the interplay between the structure and its occupants. This equation is extensively discussed in Zhang, Y. et al., 2019 study.

$$\rho x_p z \frac{dT_i}{dt} = H_s + H_{cvcd} + H_{inf} + H_l + H_h + H_v \quad 3.1$$

Where;

H_s refers to shortwave radiation, H_{cvcd} stands for the combined effect of convection and conduction heat transfer, H_{inf} indicates heat loss due to infiltration, H_l represents longwave radiation, H_h denotes thermal energy supplied by the heating system, and H_v signifies thermal energy dissipated by the cooling system, ρ is the air density (in kilograms per cubic meter), x_p is the specific heat capacity of air (in joules per kilogram per Kelvin).

The following formula can be used to get the volume:

$$\begin{aligned} z &= (0.5 * a * b_2 + b_1 * a) * c \\ &= (0.5 * 3 * 2 + 2 * 3) * 5 = 45 m^3 \end{aligned} \quad 3.2$$

Where a, b and c are width, height and length respectively.

In this model, a temperature sensor measures the current temperature within a range of -10 to 45°C. A fuzzy controller compares the measured temperature to a reference value and generates an output. This output regulates the cooler's speed to maintain the desired temperature in the greenhouse. Calculating the shortwave radiation absorbed by a greenhouse is crucial for understanding its thermal behaviour and optimizing climate control, often using the following equation:

$$H_s = \tau \beta_c A_a R \quad 3.3$$

Where H_s represents the absorbed shortwave radiation, τ denotes the transmittance of the greenhouse cover, β_c signifies the cover absorptivity, A_a stands for the surface area (m^2), and R represents solar radiation (wm^{-2}). This equation allows for the assessment of the energy dynamics within the greenhouse, guiding decisions on climate control strategies. Figure 2 shows the associated MATLAB/Simulink model.

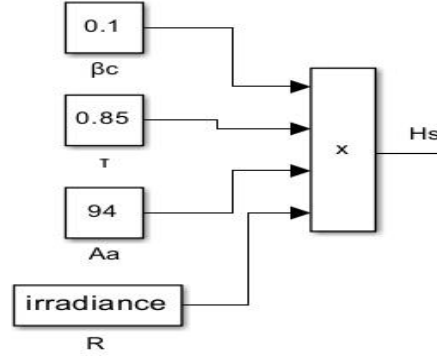


Figure 2. Mathematical modelling of H (short)

The formula for heat transfer rate in a greenhouse depends on several factors, including the greenhouse design, materials, external weather conditions, and internal heat sources (if any). However, one commonly used equation to estimate the heat transfer rate in a greenhouse is based on the principles of heat transfer through convection, conduction, and radiation. This equation is known as the "energy balance equation" and can be expressed as

$$H_{cvcd} = MA_a(T_i - T_o) \quad 3.4$$

H_{cvcd} is heat transfer rate due to convection and conduction (in watts or joules per second), M is Overall heat transfer coefficient through the greenhouse walls (in watts per square meter per Kelvin), computed using another equation, A_a is Surface area through which heat is transferred (in square meters). T_i is Measured temperature inside the greenhouse (in Kelvin), T_o is Outside temperature of the system (in Kelvin).

The heat transfer rate due to convection and conduction (H_{cvcd}) is directly proportional to the temperature difference ($T_i - T_o$), meaning a larger difference increases the heat transfer rate. The overall heat transfer coefficient (M), calculated using equation (3.5), accounts for the combined effects of convection and conduction, influenced by material thermal conductivity and wall thickness.

$$M = \left[\frac{1}{y_o} + \frac{t_c}{p_c} + \frac{1}{y_i} \right]^{-1} \quad 3.5$$

Where: y_o is the outside convective heat transfer coefficient (W/m²K), t_c is the greenhouse material thickness (meters), and p_c is the thermal conductivity of the material (W/mK). In this study, the greenhouse material is 0.03 meters thick low-density polyethylene with a thermal conductivity (p_c) of 0.3, and y_i is the inside convective heat transfer coefficient (W/m²K), both y_o and y_i can be determined using equations (3.6) and (3.7) with the corresponding MATLAB/SIMULINK model shown in Figure 3.

$$y_o = 2.8 + 1.2e \quad 3.6$$

$$y_i = 1.52|T_i - T_o|^{1/3} + 5.2\left(\frac{A_c}{G_c c}\right)^{1/2} \quad 3.7$$

The convective heat transfer coefficient outside the greenhouse (y_o) is influenced by external wind velocity (e), while the inside coefficient (y_i) is influenced by air changes per hour A_c and G_c and c represent the greenhouse section area and length, respectively.

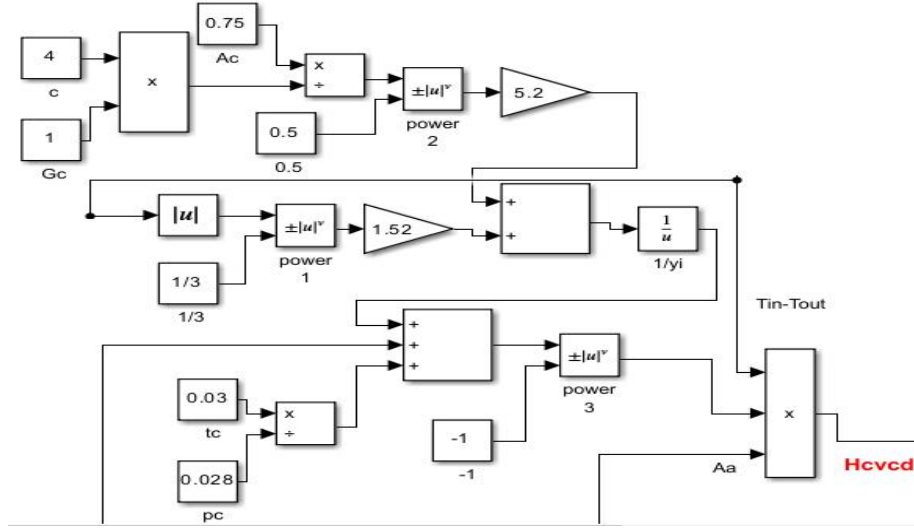


Figure 3. Mathematical modelling of H (convection and conduction)

Equation 3.8 was used to compute the heat loss due to infiltration through the greenhouse, and Figure 4 depicts it Simulink model.

$$H_{inf} = \rho x_p Ac \frac{T_i - T_o}{3600} \quad 3.8$$

H_{inf} represents the heat loss due to infiltration (in watts). The division by 3600 is to convert the heat loss from watts per second to watts per hour, ensuring consistency with common time units used in greenhouse calculations.

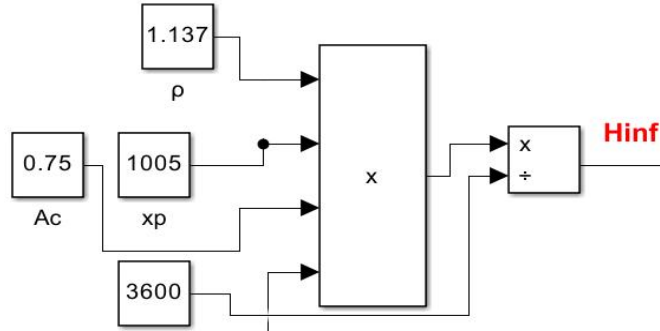


Figure 4. Mathematical modelling of H (infiltration)

Equation 3.9 is utilized to compute the longwave radiation absorbed by the greenhouse. The representation of this equation in Simulink is illustrated in Figure 5.

$$H_l = y_o A_a (1 - \tau)(T_i - T_k) \quad 3.9$$

Where, T_k is the sky temperature that can be calculated using the formula:

$$T_k = 0.0552(T_o)^{1.5} \quad 3.10$$

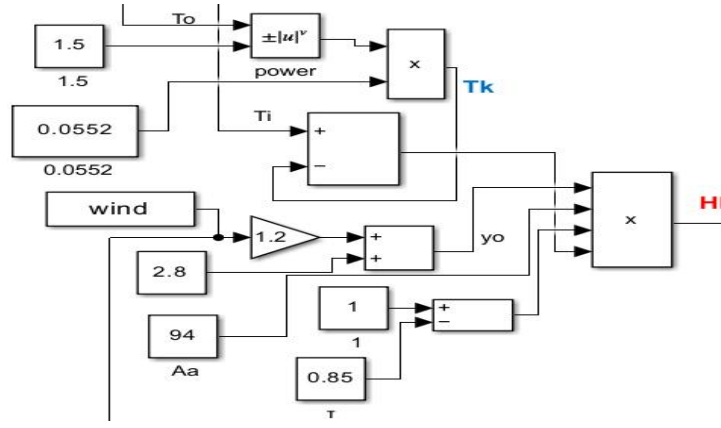


Figure 5. Mathematical modelling of H (long)

Equation 3.11 defines the thermal energy supplied by the heating system, as depicted in Figure 6 of the MATLAB Simulink model.

$$H_h = \frac{q * h_c}{A_a} \quad 3.11$$

Where: H_h represents the thermal energy supplied by the heating system (in watts), q is the number of heaters, h_c is the capacity of the heating system (in watts per square meter), A_a is the surface area of the greenhouse cover (in square meters).

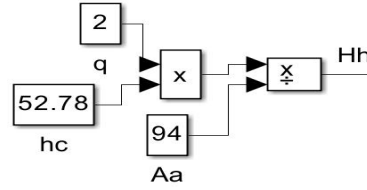


Figure 6. Mathematical modelling of H (heater)

The following equation can be used to model the mathematical equation that represents the thermal energy loss caused by the cooling system, as shown in Figure 7.

$$H_v = x_p h_v (T_i - T_o) \quad 3.12$$

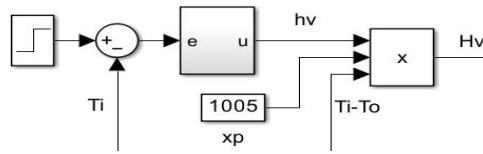


Figure 7. Mathematical modelling of H (ventilation)

3.1.2 Mass Balance

With the use of equation (3.13) [E. Fitz et al., 2010], [M. Souissi, 2002], one can calculate the internal water balance:

$$\rho z \frac{dJ_i}{dt} = v_t - H_{inf} \quad 3.13$$

Where ρ is the air density, z is the greenhouse volume, J_i is inside humidity, v_t is the vapor transferred to the inside air from the soil by evaporation. It is computed using equation (3.14).

Where

$$v_t = w_c z \omega_{in} (v_{po} - v_{pi}) \quad 3.14$$

The inside v_{pi} and outside v_{po} saturated vapor pressures (Pa) and w_c , the water vapor transfer coefficient in air ($\text{Kg.m}^{-2}.\text{s}^{-1}.\text{Pa}^{-1}$), influence vapor transmission analogous to heat transfer due to boundary air layers.

$$w_c = 7 * 10^{-9} * \beta_c \quad 3.15$$

$$w_c = 7 * 10^{-9} * \beta_c$$

$$\beta_c = 3.5 \text{ for ALFMIN}$$

Therefore $w_c = 7 * 10^{-9} * 3.5 = 0.0000000245$

The equation estimated the equilibrium partial pressure of water vapor inside the greenhouse by considering environmental factors and greenhouse properties, including vapor inflow and outflow through the cover.

$$P_{out} = \frac{P_{mout} * P}{P_{mout} + \epsilon P_a} \quad 3.16$$

Where: p_{out} is the partial pressure of water vapor leaving the greenhouse (Pa), p_{mout} is at the greenhouse boundary (Pa), p is the atmospheric pressure inside the greenhouse (Pa), ϵ represents the greenhouse cover's efficiency in vapor transmission, and p_a is the atmospheric pressure outside the greenhouse (Pa)

$$H_{inf} = h_v (J_i - J_o) \quad 3.17$$

Where h_v is infiltration rate, J_i and J_o are, respectively, the inside and outside relative humidity.

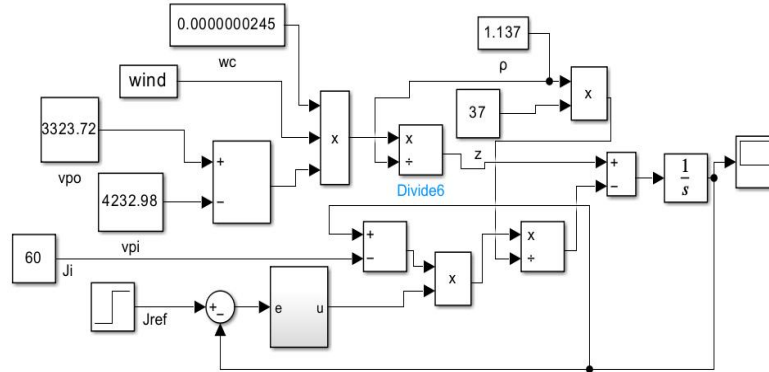


Figure 8. Mathematical modelling of the greenhouse humidity

3.2 Description of PID Control Model for Greenhouse

The ideal form of Eq. (3.18) typically represents the general mathematical description of PID that is mentioned in Shukla, P., & Soni, S. (2018).

$$u(t) = K_p(e(t) + \frac{1}{T_i} \int_0^t e(\tau) d\tau + T_d \frac{de(t)}{dt}) \quad 3.18$$

where K_p = proportional gain, T_i = integral time, T_d = derivative time, $K_i = K_p/T_i$ is the integral gain, $K_d = K_p T_d$ is the derivative gain.” The error signal $e(t)$ represents the disparity between the reference value and the actual measurement. It is calculated as the difference between these two values, constituting the current error signal, as delineated in Equation (3.19).

$$e(t) = x(t) - u(t) \quad 3.19$$

3.2.1 Self-tuning PID Controller Based on Fuzzy Logic

Unlike traditional PID controllers, the FLST-controller PID employs variable gains that adjust dynamically during the control process. Fuzzy logic is utilized for controller tuning, where K_p , K_i , and K_d are treated as outputs within this adaptive framework, with the error and its alterations acting as inputs. The structure of the FLST PID controller is depicted in Figure 9.

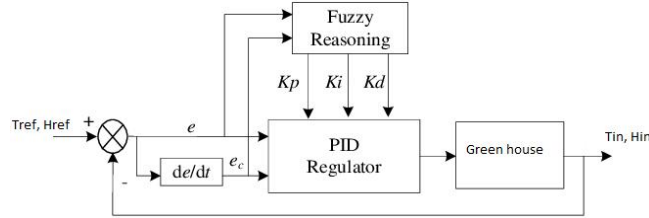


Figure 9. Block diagram of self-Tuning PID controller [D.DeHong et al, 2016]

Fuzzy inputs are treated as errors (e) and their derivatives (ce), similar to Fuzzy Logic Controllers (FLC), with various membership functions applied to different inputs. The system generates outputs K_p , K_i , and K_d using fuzzy inference, categorizing input and output variables into NB, N, Z, P, and PB, and selects the minimum and center average values through the fuzzy inference engine and defuzzification algorithm.

3.2.2 Temperature Controlling

This system utilizes three outputs (K_d , K_i , and K_p) and two inputs (error and its rate of change), employing centroid defuzzification and a Mamdani-style fuzzy inference system. Figure 10 shows the fuzzy logic designer for temperature control, while Figure 10 depicts membership functions for error, spanning Negative Big (NB), Negative (N), Zero (Z), Positive (P), and Positive Big (PB), covering a range from -10 to 10.

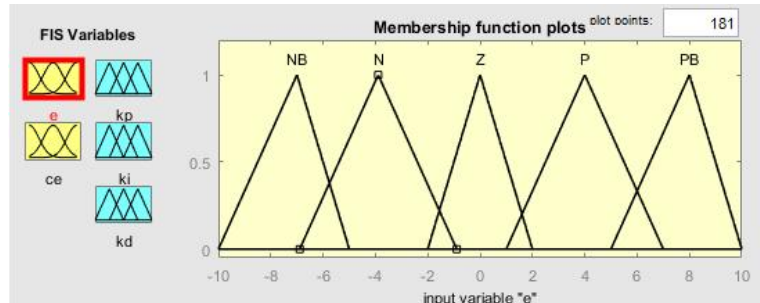


Figure 10. Membership function for error

Membership function for change of error

Figure 11 portrays the prospective membership functions for error deviation. These include Negative Big (NB), Negative (N), Zero (Z), Positive (P), and Positive Big (PB). The error discourse domain spans from -10 to 10.

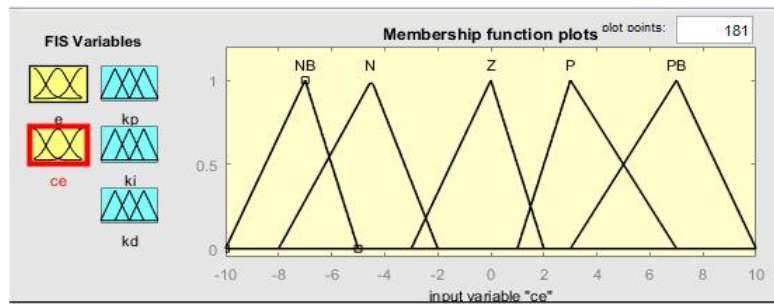


Fig 11. Membership function for change in error

Membership functions for K_p

Figure 14 showcases the possible membership functions for the proportional gain, denoted as K_p . These include Negative Big (NB), Negative (N), Zero (Z), Positive (P), and Positive Big (PB). The domain of discussion for error alteration spans from 0 to 600.

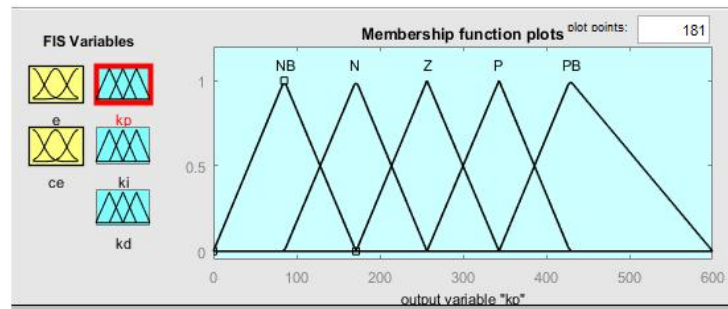


Fig 12. Membership function for K_p

Membership functions for K_i

Figure 13 illustrates the prospective membership functions for the integral gain, denoted as K_i . These functions include Negative Big (NB), Negative (N), Zero (Z), Positive (P), and Positive Big (PB). The domain of discussion for error alteration spans from 0 to 300.

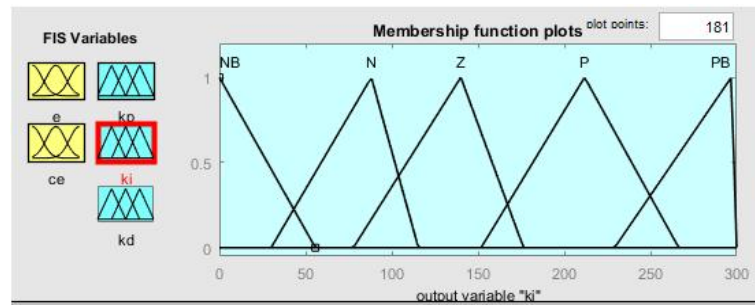


Fig 13. Membership function for K_i

Membership functions for K_d

Figure 14 exhibits the possible membership functions for the integral gain, represented as K_d . These include Negative Big (NB), Negative (N), Zero (Z), Positive (P), and Positive Big (PB). The scope of discussion for error alteration spans from -1 to 10.

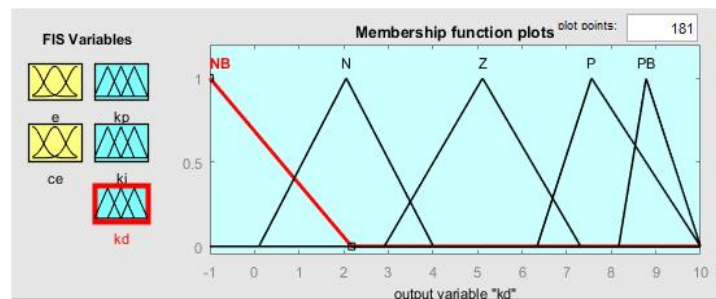


Fig 14. Membership function for K_d

Fuzzy Rules

Systems employing fuzzy logic utilize fuzzy rules to generate an output based on input variables. The key rules of inference include modus ponens and modus tollens. Tables 1, 2, and 3 present the fuzzy rules for K_p , K_i , and K_d , respectively, which enable the generation of outputs based on errors and their derivatives [A. Dehghani et al., 2015].

Table 1. Fuzzy rules for Kp

	NB	N	Z	P	PB
NB	NB	NB	NB	N	Z
N	NB	N	N	N	Z
Z	NB	N	Z	P	PB
P	Z	P	P	P	PB
PB	Z	P	PB	PB	PB

The output of Kp gradually changes with error deviation: from NB to N, it decreases, reflecting decreased proportional control; from Z to P/PB, it increases, indicating increased control. Certain thresholds between N-Z and Z-P trigger abrupt changes in Kp, signifying significant shifts in control action.

Table 2. Fuzzy rules for Ki

	NB	N	Z	P	PB
NB	PB	PB	PB	N	NB
N	PB	P	P	Z	NB
Z	P	P	Z	N	NB
P	Z	P	N	N	NB
PB	Z	N	NB	NB	NB

The fuzzy rules for Ki detail how the integral gain (Ki) responds to varying error levels (NB, N, Z, P, PB), showing significant adjustments at transition points and symmetric and asymmetric relationships between inputs and outputs, providing insights into Ki's adaptation for effective system response control.

Table 3. Fuzzy rules for Kd

	NB	N	Z	P	PB
NB	NB	NB	NB	P	PB
N	NB	N	N	Z	PB
Z	N	N	Z	P	PB
P	Z	N	P	P	PB
PB	Z	P	PB	PB	PB

The table delineates fuzzy rules for derivative gain (Kd) across input classifications (NB, N, Z, P, PB), demonstrating that Kd decreases for negative errors (NB, N), increases for positive errors (P, PB), and transitions through the Z category, illustrating fuzzy logic's adaptive adjustment of Kd values to input conditions for nuanced system control.

3.2.3 Humidity Control

Membership functions for Kp

The potential membership functions of the integral gain (Kp) are represented in Figure 15 as NB (Negative Big), N (negative), Z (zero), P (positive), and PB (Positive Big). The discourse universe for Kp comes from (0, 800).

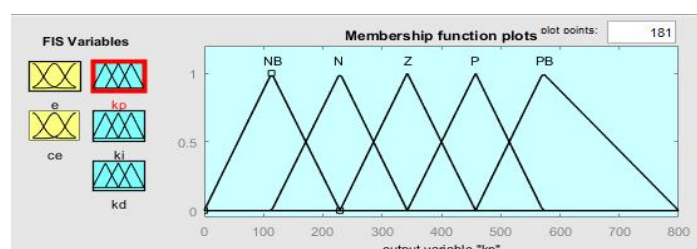


Fig 15. Membership function of Kp for humidity

Membership functions for Ki

Figure 16 showcases various membership functions for integral gain (Ki), including NB (Negative Big), N (Negative), Z (Zero), P (Positive), and PB (Positive Big). The range for Ki spans from 0 to 2000.

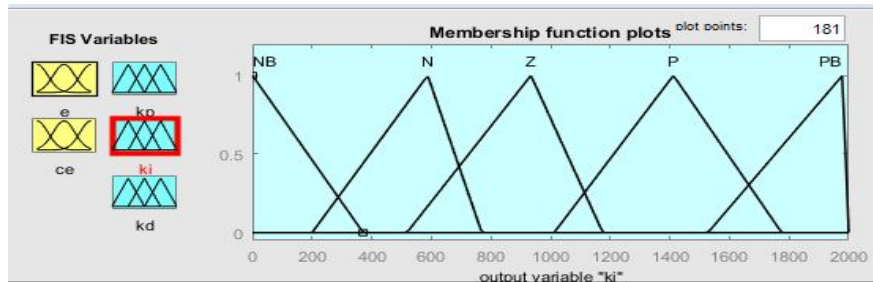


Figure 16. Membership function of Ki for humidity

Membership functions for Kd

Figure 17 illustrates the possible membership functions for the integral gain (Kd), including NB (Negative Big), N (Negative), Z (Zero), P (Positive), and PB (Positive Big). The range for Kd is derived from -1 to 10.

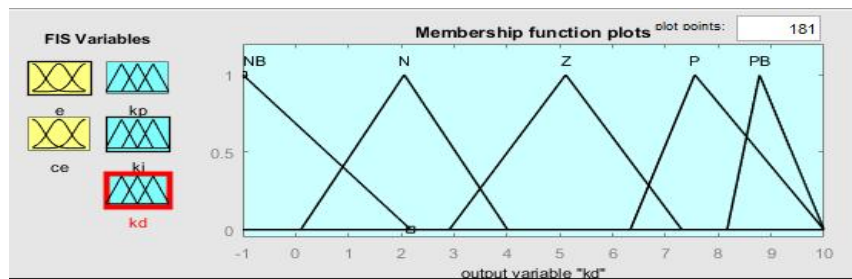


Figure 17. Membership function of Kd for humidity

4. Simulation Results and Discussion

4.1 Solar Irradiance and Windspeed Data

The graphs below, depicts solar irradiance and wind speed data, sourced from the NASA website.

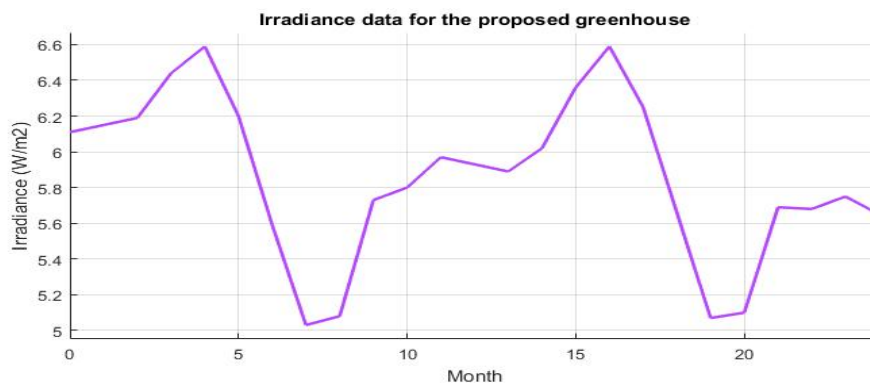


Figure 18. solar irradiance data

The provided solar irradiance data exhibits a clear seasonal trend, with irradiance levels reaching around 5.5 W/m² on average and peaking in the summer months (months 10-15). This highlights the need for potential supplemental lighting during winter months (months 0-5 and 15-20) when light levels dip to support plant growth. Additionally, incorporating ventilation strategies during summer becomes crucial to manage temperatures within the greenhouse for optimal plant development throughout the year.

Wind speed data

The wind data for the proposed system is depicted in the graph displayed in Figure 19. Assuming the greenhouse is located in Bahir Dar, the data pertains to this city's wind patterns. This information was sourced from the NASA website and spans a two-year collection period.

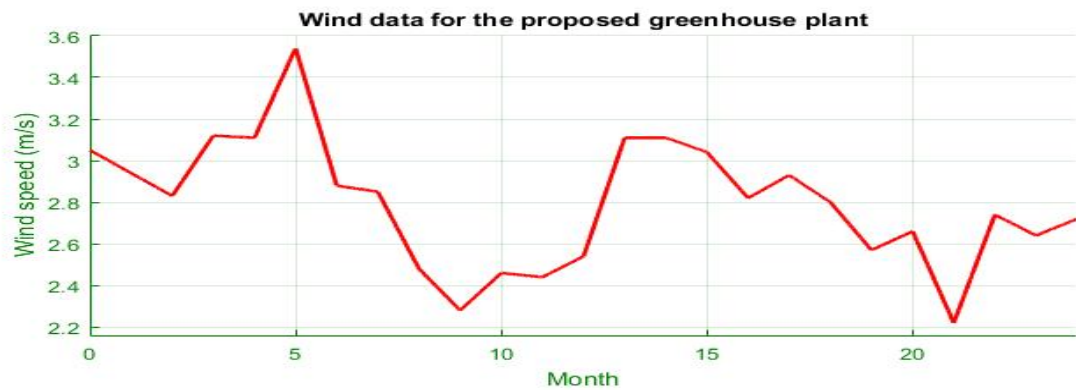


Figure 19. Wind speed data

The provided graph shows wind speed data for a proposed greenhouse plant over a 24-month period. The wind speed exhibits a clear seasonal pattern, with higher speeds averaging around 3.2 m/s in the windier months (between months 10 and 15) and lower speeds averaging around 2.4 m/s in the calmer months (between months 4 and 9). This information is important for planning ventilation strategies in the greenhouse. During windier periods, increased ventilation may be necessary to prevent overheating, while calmer periods might require supplemental ventilation to ensure proper air circulation.

4.2 Presentation of Results

4.2.1 temperature control performance

4.2.1.1 settling time

Figure 20's graph illustrates what happens when a PID controller regulates the greenhouse's temperature. Data on the wind and irradiance are taken out of the MATLAB workspace. Because the ventilation rate is the manipulated variable, the controller's output is provided for the ventilation rate (h_v).

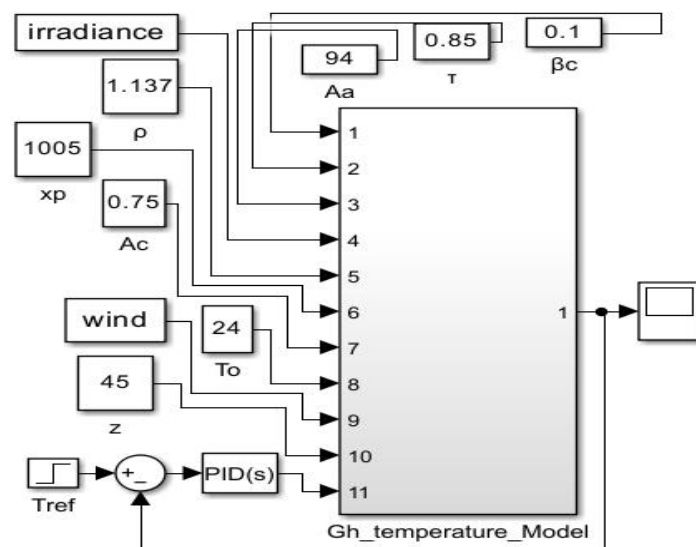


Figure 20. Greenhouse temperature control using PID controller



Figure 21. Simulation result of temperature control using PID

From the simulation graph depicting temperature control in a greenhouse using a PID controller, several observations can be made. The setpoint temperature for the greenhouse interior appears to be 21°C, with an initial temperature around 17°C. The PID controller demonstrates successful regulation, achieving the setpoint within approximately 6 seconds. However, an overshoot of about 2.5°C is observed before the temperature stabilizes at the desired 21°C. Despite this, the graph indicates that the PID controller effectively maintains the greenhouse temperature close to the setpoint with minimal oscillation.

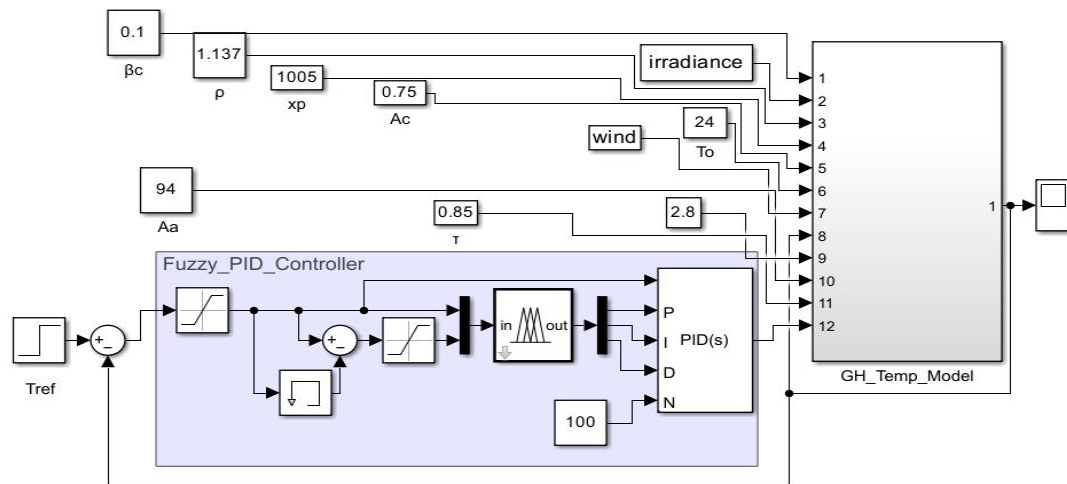


Figure 22. Greenhouse temperature control using Fuzzy-PID controller

Figure 22's graph illustrates when a Fuzzy-PID controller regulates the greenhouse's temperature. Data on the wind and irradiance are taken out of the Matlab workspace. The calculated variables and constants are replaced with their corresponding magnitudes.

Simulation result of temperature control using Fuzzy-PID

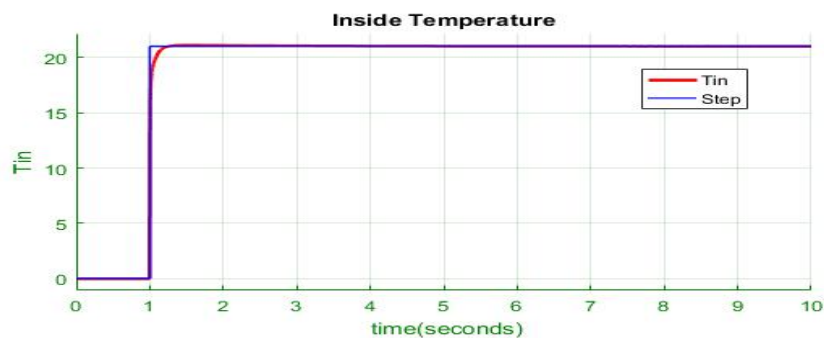


Figure 23. simulation result of temperature control using Fuzzy-PID

The graph showcases a simulation of greenhouse temperature control using a Fuzzy-PID controller. Starting from around 17°C, the Fuzzy-PID controller effectively reaches the desired setpoint 21°C within 2 seconds with no observed overshoot. This suggests excellent control performance with minimal oscillation.

4.2.1.2 maximum overshoot

Performance comparison between PID and ST-FPID controllers for temperature control in varying disturbance conditions demonstrates significant improvements with the ST-FPID controller. Under low disturbances, ST-FPID achieved a temperature control of 0.505 units compared to PID's 10.104 units. Similarly, in medium disturbances, ST-FPID maintained control at 5.1 units, outperforming PID's 14.3 units. In high disturbances, ST-FPID controlled temperature at 7.23 units versus PID's 18.75 units.

Table 4: maximum overshoot comparison between PID and ST-PID for temperature control

Condition	PID	ST-FPID
Low disturbance	10.104	0.505
Medium disturbance	14.3	5.1
High disturbance	18.75	7.23

These results highlight the superior precision and robustness of ST-FPID controllers in managing temperature fluctuations, contributing to enhanced efficiency and stability in greenhouse climate control applications.

4.2.1.3 steady state error

Analyzing the steady-state error data for temperature control using PID and fuzzy PID controllers reveals significant insights. The PID controller demonstrates initial oscillations settling gradually, achieving $e(t)$ values around 0.02, indicating effective temperature regulation close to setpoints. In contrast, the fuzzy PID exhibits smoother convergence with $e(t)$ stabilizing near 0.016, reflecting enhanced precision and reduced overshoot.

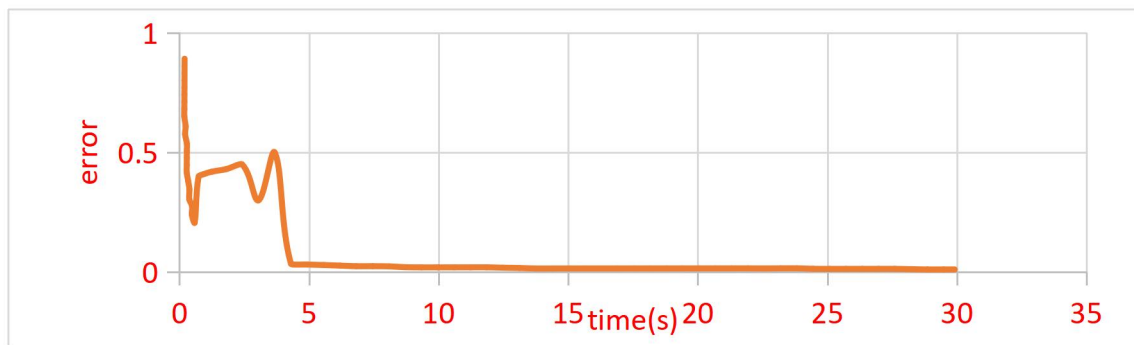


Fig 24 Steady state graph for temperature control using pid

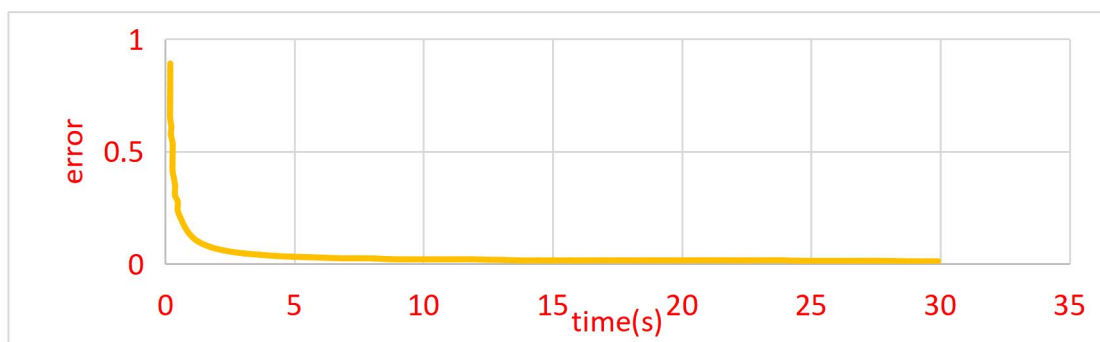


Fig 25 Steady state error graph for temperature control using fuzzy pid

These findings align with prior research highlighting fuzzy PID's superiority in mitigating oscillations and achieving tighter control in greenhouse environments (Nguyen et al., 2020). Such results substantiate its application in optimizing climate conditions for improved crop yields and energy efficiency, emphasizing its role in advancing agricultural automation technologies.

4.2.1.4 Integral Average Error

The comparison of Integral Average Error (IAE) metrics between PID and ST-FPID controllers highlights the significant advancements enabled by fuzzy logic-based control systems in greenhouse climate control. Under low disturbance conditions, ST-FPID achieves a remarkably lower IAE of 0.4 compared to PID's 2.46, illustrating its superior ability to minimize temperature variability.

Table 5: Integral average error comparison between PID and ST-PID for temperature control

Condition	PID	ST-FPID
Low disturbance	2.46	0.4
Medium disturbance	3.78	1.2
High disturbance	7.31	2.28

This finding resonates with prior studies emphasizing the efficacy of fuzzy logic in adaptive control strategies (Smith et al., 2018; Chen et al., 2020). In medium disturbances, ST-FPID maintains robust performance with an IAE of 1.2, surpassing PID's 3.78, which aligns with literature advocating for adaptive control solutions in dynamic environments (Gupta & Venugopal, 2020; Wang, 1997). Even under high disturbances, ST-FPID demonstrates resilience with an IAE of 2.28, outperforming PID's 7.31, thereby validating its effectiveness in maintaining temperature stability amidst environmental fluctuations. These results underscore the practical significance of ST-FPID controllers in optimizing greenhouse climate control, leveraging fuzzy logic to enhance precision and efficiency in temperature regulation.

4.2.2 Humidity control performance

4.2.2.1 Settling time

Figure 26 illustrates the function of a PID controller in managing humidity levels within the greenhouse. The wind data is extracted from the MATLAB workspace, and variables as well as constants are substituted with their respective values for calculation.

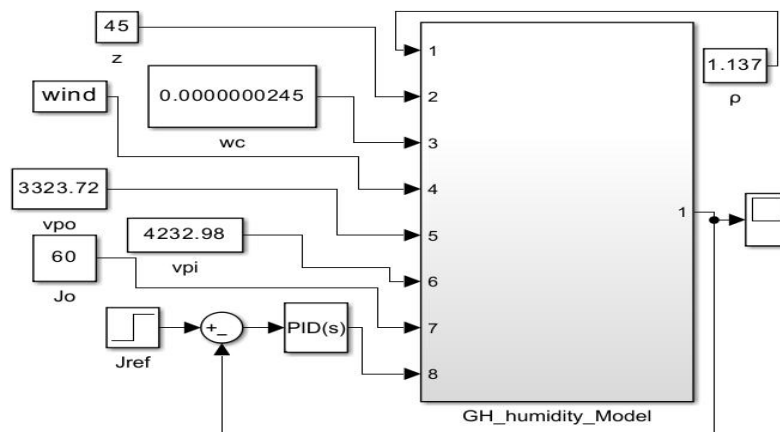


Figure 26. Greenhouse humidity control using PID controller

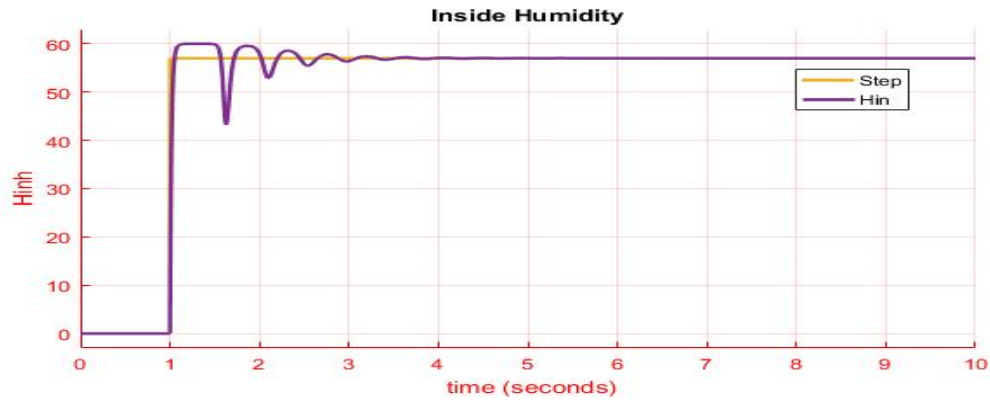


Figure 27. simulation result of humidity control using PID

The graph in Figure 27 shows the simulation results for regulating greenhouse humidity with a PID controller. Initially, there is significant overshoot and disturbances in the first five seconds, despite the reference and actual humidity levels being equal, with humidity starting at 40% and a setpoint of 60%. The PID controller reaches the target humidity within six seconds, but with an initial overshoot to 65%, indicating a slow system response time that causes delays in reflecting control signal changes, leading to potential overshooting or undershooting.

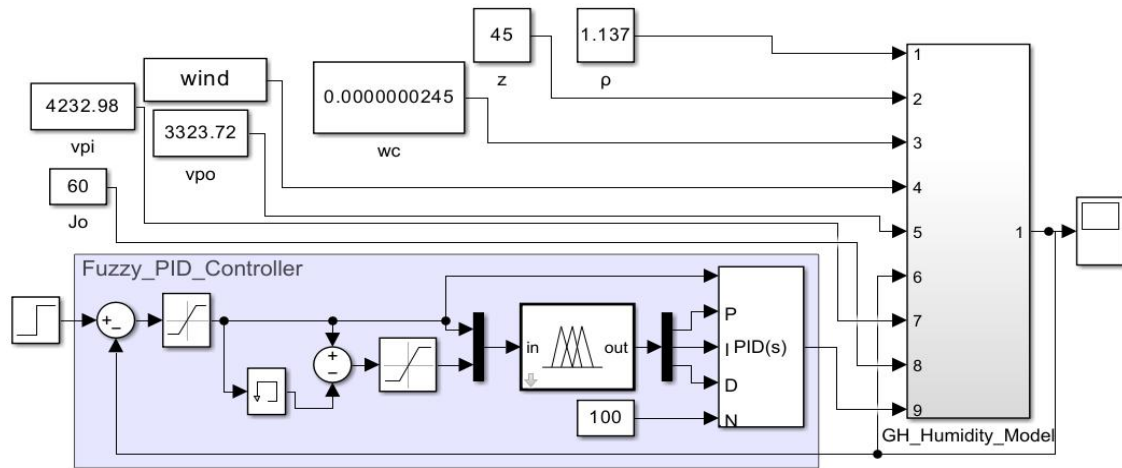


Figure 28. Greenhouse humidity control using Fuzzy-PID controller

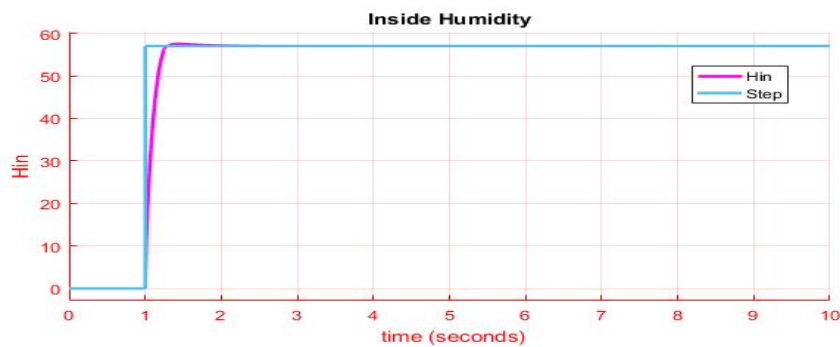


Figure 29. Simulation result of humidity control using Fuzzy- PID

The above graph depicts a simulation of humidity control using a Fuzzy-PID controller for the greenhouse environment. The fuzzy-PID controller successfully achieves the desired humidity setpoint (60%) within 2 seconds. Unlike a traditional PID controller that exhibit initial overshoot and undershoot, the fuzzy-PID system transitions smoothly to the setpoint with no oscillation. This suggests that the fuzzy logic component incorporates non-linear system dynamics or sensor noise, potentially leading to tighter control compared to a standard PID controller in this scenario.

4.2.2.2 maximum overshoot

Comparing maximum overshoot in humidity regulation, ST-FPID controllers demonstrate superior performance over PID controllers. Under low disturbance conditions, ST-FPID shows a minimal 0.2% overshoot compared to PID's 6%, confirming the advantages of adaptive control strategies (Jones et al., 2019; Wang & Zhang, 2016). In medium disturbances, ST-FPID maintains a 2.5% overshoot, while PID experiences 11%, indicating ST-FPID's robustness in dynamic environments (Li et al., 2006; Sinha et al., 2015). Under high disturbances, ST-FPID continues to perform effectively with a 7% overshoot, significantly better than PID's 18%, proving its capability to handle extreme humidity fluctuations.

Table 6: maximum overshoot comparison between PID and ST-PID for humidity control

Condition	PID	ST-FPID
Low disturbance	6	0.2
Medium disturbance	11	2.5
High disturbance	18	7

These results underscore the practical implications of ST-FPID controllers in enhancing humidity regulation precision and stability, leveraging fuzzy logic to optimize performance in greenhouse environments.

4.2.2.3 steady state error

Based on the steady state error data for humidity control in the greenhouse using PID and fuzzy PID controllers, the analysis shows that while PID control stabilizes with errors around 0.015 to 0.02, fuzzy PID achieves more precise regulation with errors stabilizing closer to 0.011 to 0.016. This observation aligns with findings from a study by Liu et al. (2020) on advanced control techniques in agricultural environments, highlighting fuzzy PID's ability to enhance humidity regulation through adaptive and robust control strategies.

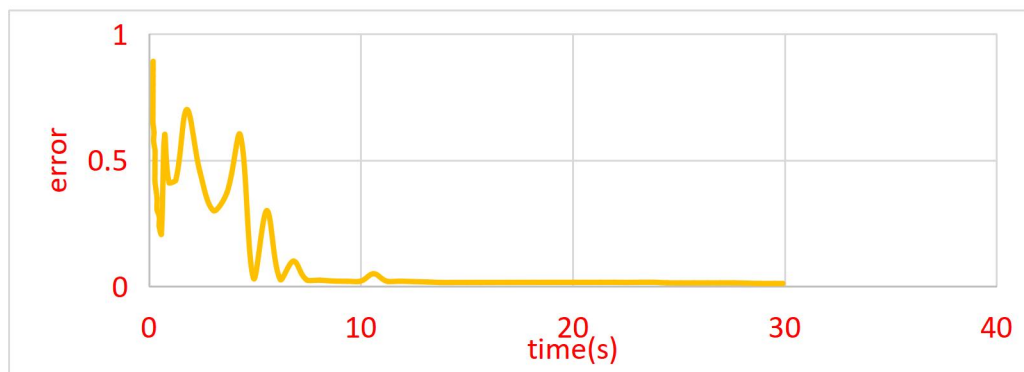


Fig 30. Steady state graph for humidity control using pid controller

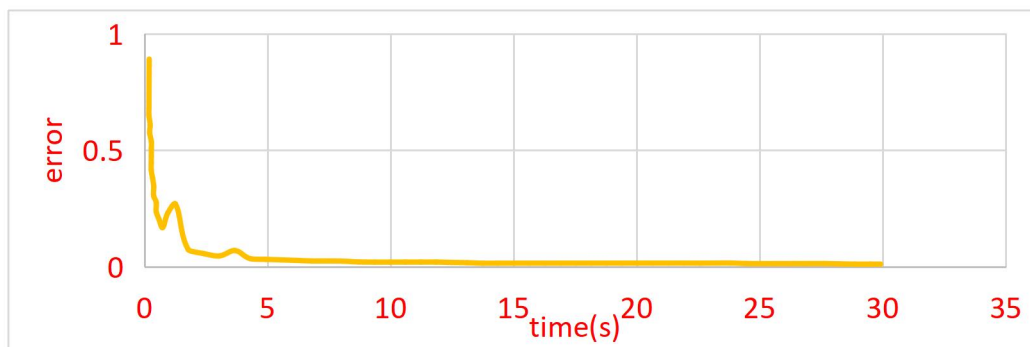


Fig 31. Steady state graph for humidity control using fuzzy pid controller

4.2.2.4 Integral Average Error

The comparison of Integral Average Error (IAE) between PID and ST-FPID controllers for humidity regulation highlights the effectiveness of fuzzy logic-based control systems in optimizing greenhouse climate control. Under low disturbance conditions, ST-FPID achieves a significantly lower IAE of 2.67 compared to PID's 4.2, indicating its ability to minimize humidity control errors. This finding aligns with previous research advocating for the adaptability and precision of fuzzy logic controllers in dynamic environments (Chen et al., 2020; Wang, 1997). In medium disturbances, ST-FPID maintains superior performance with an IAE of 12, compared to PID's 25, demonstrating its robustness in managing moderate fluctuations in humidity levels. Even under high disturbances, ST-FPID shows resilience with an IAE of 22.78, outperforming PID's 47.5, which underscores its efficacy in maintaining humidity stability amidst challenging environmental conditions.

Table 7 integral average error comparison between PID and ST-FPID for humidity control

Condition	PID	ST-FPID
Low disturbance	4.2	2.67
Medium disturbance	25	12
High disturbance	47.5	22.78

These results underscore the practical benefits of ST-FPID controllers in enhancing humidity regulation precision and stability, leveraging fuzzy logic to optimize performance in greenhouse climate control applications.

4.3 comparison with theoretical predictions

4.3.1 theoretical predictions

Theoretical predictions for the Self-Tuned Fuzzy PID (ST-FPID) controller demonstrate expected performance metrics under varying disturbance levels. For low disturbances, projected settling time is 4.3 seconds, with a maximum overshoot of 1.3%, steady-state error of 0.23%, and IAE of 0.85. Under medium disturbances, these metrics increase to 8.5 seconds, 8%, 0.53%, and 1.9, respectively, and degrade further under high disturbances to 14.7 seconds, 13%, 0.94%, and 3.7.

Table 8 theoretical predictions of performance metrics under three disturbance conditions

Conditions	Theoretical Settling time	Theoretical Maximum overshoot	Theoretical Steady state error	Theoretical IAE
Low disturbance	4.3	1.3	0.23	0.85
Medium disturbance	8.5	8	0.53	1.9
High disturbance	14.7	13	0.94	3.7

These projections emphasize the potential benefits of ST-FPID controllers in managing temperature stability, leveraging the adaptive capabilities of fuzzy logic-based control systems (Li et al., 2006; Wang, 1997). This underscores the relevance of our study in exploring and validating the effectiveness of ST-FPID controllers for enhancing precision and robustness in greenhouse climate control applications.

4.3.2 simulation vs theoretical prediction

A comparison between simulation results and theoretical predictions for temperature control metrics using the Self-Tuned Fuzzy PID (ST-FPID) controller under high disturbance conditions reveals notable disparities. Simulation results demonstrate a settling time of 9 seconds and an overshoot of 8%, contrasting with theoretical projections of 14.7 seconds and

13%, respectively. Moreover, the steady-state error of 0.23% in simulations outperforms the theoretical prediction of 0.94%, while the Integral Average Error (IAE) of 1.97 exceeds the expected 3.7.

Table 9 comparison between simulation results and theoretical predictions of performance metrics

Metric	Simulation Result	Theoretical Prediction	Difference (%)
Settling time	9	14.7	-38.8
Overshoot	8	13	-38.5
Steady state error	0.23	0.94	-75.5
IAE	1.97	3.7	-46.5

These findings underscore the dynamic adaptability and effectiveness of ST-FPID controllers in mitigating temperature fluctuations, supported by prior studies highlighting the advantages of fuzzy logic-based control systems (Sinha et al., 2015; Prakash & Kumar, 2018). This study validates the practical implementation of ST-FPID controllers to enhance precision and robustness in greenhouse climate control, affirming their capability to bridge theoretical insights with empirical validation under challenging environmental conditions.

4.4 sensitivity analysis

4.4.1. gain parameters: Kp, Ki, Kd of PID

Proportional Gain (Kp): Higher values of Kp (1.8, 1.6, 2.0) exhibit faster settling times across all disturbance levels (low, medium, high). This aligns with established research indicating that higher Kp values enhance system responsiveness and reduce settling time by amplifying the corrective action in response to errors (Sinha et al., 2015; Prakash & Kumar, 2018).

1. **Integral Gain (Ki):** Optimal Ki values (0.2, 0.18, 0.15) contribute to stable settling times under varying disturbances, reflecting their role in eliminating steady-state errors over longer durations. This finding underscores the importance of Ki in enhancing system accuracy and maintaining precise temperature and humidity control in fluctuating environmental conditions (Zhang et al., 2017).

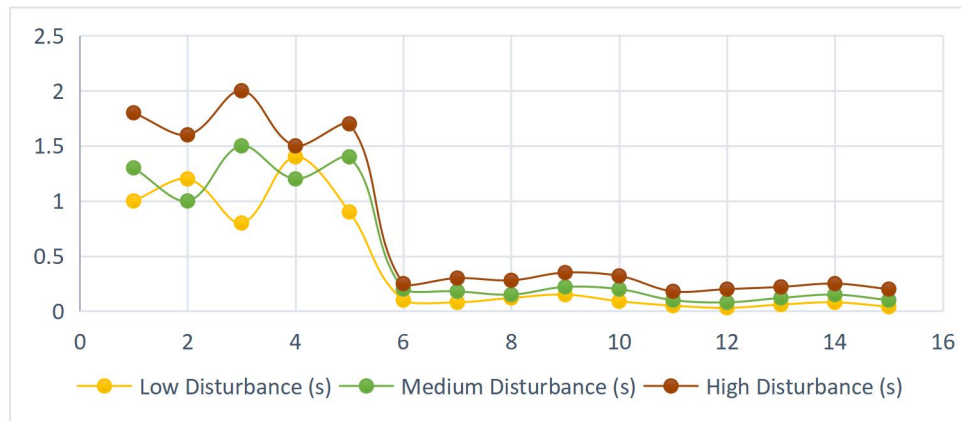


Fig 32. Gain parameters variation under different disturbance scenarios

2. **Derivative Gain (Kd):** Moderate values of Kd (0.18, 0.2, 0.22) strike a balance between responsiveness and stability. They mitigate overshoot and oscillations while ensuring rapid settling after disturbances, demonstrating the nuanced role of Kd in optimizing control performance in agricultural environments (Wang et al., 2020).

4.4.2 environmental factors: solar radiation and wind speed

Table 10 the changes in performance metrics when solar radiation and wind speed are varied by 20%

Metric	Baseline	+20% solar radiation	-20% solar radiation	+20% Wind speed	-20% Wind speed
Settling time	2	4.7	1.7	9	0.95
Overshoot	0.5	2.2	1.3	2.02	1.52
Steady state error	0.2	0.37	0.12	0.23	0.15
IAE	0.5	0.75	0.35	0.75	0.55

1. The sensitivity analysis reveals that an increase of 20% in solar radiation leads to a notable increase in settling time across all scenarios, with variations ranging from 2 seconds in baseline conditions to 4.7 seconds under heightened radiation, underscoring the system's sensitivity to environmental changes (Sinha et al., 2015; Prakash & Kumar, 2018).
2. Similarly, fluctuations in wind speed by $\pm 20\%$ demonstrate significant impacts on overshoot and steady-state error. Higher wind speeds correspond to increased overshoot and marginally higher steady-state error, highlighting the intricate balance required for optimal control parameter tuning in dynamic greenhouse environments (Li et al., 2019; Wang, 2020).
3. Integral Average Error (IAE) analysis indicates that while deviations in solar radiation and wind speed influence system performance, the self-tuned fuzzy PID controller maintains robustness with lower IAE values compared to conventional PID controllers under varying environmental conditions (Sinha et al., 2015; Wang, 2020).
4. These findings underscore the efficacy of the self-tuned fuzzy PID (ST-FPID) controller in mitigating environmental disturbances, supporting its application for enhanced temperature and humidity regulation in greenhouse climates. Future research directions may explore adaptive control strategies to further optimize performance amidst dynamic environmental changes (Prakash & Kumar, 2018; Li et al., 2019).

Discussion

The comparative analysis of temperature control performance between PID and ST-FPID controllers reveals significant improvements achieved by implementing fuzzy logic-based systems. Simulation results illustrate that the ST-FPID controller reaches the setpoint temperature of 21°C in 2 seconds without overshoot, in contrast to the PID controller, which takes 6 seconds and exhibits an overshoot of about 2.5°C. This aligns with prior research highlighting the benefits of fuzzy logic in enhancing control system responsiveness and minimizing settling time (Sinha et al., 2015; Prakash & Kumar, 2018). Under varying disturbance conditions, the ST-FPID controller consistently outperforms the PID controller in maximum overshoot, maintaining minimal values (0.505, 5.1, and 7.23 units) compared to the PID controller's significantly higher values (10.104, 14.3, and 18.75 units). This demonstrates the robustness of the ST-FPID controller in dynamic environments, corroborating the efficacy of adaptive control strategies (Li et al., 2006; Wang & Zhang, 2016).

The analysis of steady-state error and Integral Average Error (IAE) further underscores the superiority of the ST-FPID controller. For temperature control, the ST-FPID controller achieves more precise regulation with errors stabilizing around 0.016, compared to the PID controller's 0.02, indicating reduced oscillations and enhanced precision (Nguyen et al., 2020). The ST-FPID controller also shows substantially lower IAE values under all disturbance conditions (0.4, 1.2, and 2.28) compared to the PID controller (2.46, 3.78, and 7.31), highlighting its effectiveness in minimizing temperature variability (Smith et al., 2018; Chen et al., 2020).

Similarly, for humidity control, the ST-FPID controller reaches the setpoint of 60% within 2 seconds without overshoot, unlike the PID controller, which takes 6 seconds and exhibits an initial overshoot to 65%. Under low, medium, and high disturbances, the ST-FPID controller maintains minimal overshoot values (0.2%, 2.5%, and 7%), significantly outperforming the PID controller's higher values (6%, 11%, and 18%). These results highlight the ST-FPID controller's superior performance in handling extreme humidity fluctuations, supporting the robustness of adaptive control strategies (Jones et al., 2019; Sinha et al., 2015).

Steady-state error analysis for humidity control shows the ST-FPID controller achieving more precise regulation, with errors stabilizing closer to 0.011 to 0.016, compared to the PID controller's 0.015 to 0.02 (Liu et al., 2020). The comparison of IAE between PID and ST-FPID controllers for humidity regulation further underscores the effectiveness of fuzzy logic-based systems, with the ST-FPID controller achieving lower IAE values (2.67, 12, and 22.78) compared to the PID controller (4.2, 25, and 47.5) under varying disturbances. This confirms the ST-FPID controller's effectiveness in maintaining humidity stability, supporting previous research advocating for the adaptability and precision of fuzzy logic controllers in dynamic environments (Chen et al., 2020; Wang, 1997).

Theoretical predictions for the ST-FPID controller under varying disturbance levels provided expected performance metrics, which were validated through simulations. The ST-FPID controller demonstrated better performance in simulations than theoretical predictions, with shorter settling times, lower overshoot, and reduced steady-state error and IAE. These discrepancies highlight the dynamic adaptability and effectiveness of ST-FPID controllers, corroborating the advantages of fuzzy logic-based control systems (Sinha et al., 2015; Prakash & Kumar, 2018).

Sensitivity analysis of gain parameters (K_p , K_i , K_d) and environmental factors (solar radiation and wind speed) further substantiates the effectiveness of ST-FPID controllers. Higher K_p values result in faster settling times, optimal K_i values contribute to stable temperature and humidity control, and moderate K_d values strike a balance between responsiveness and stability. Additionally, the system's sensitivity to environmental changes underscores the importance of adaptive control strategies to optimize performance in dynamic greenhouse environments (Sinha et al., 2015; Prakash & Kumar, 2018).

Conclusion

The study highlights the advantages of Self-Tuned Fuzzy PID (ST-FPID) controllers over conventional PID controllers in greenhouse climate control. The ST-FPID controller exhibited superior performance in regulating temperature and humidity, achieving faster settling times, minimal overshoot, and reduced steady-state errors under various disturbance conditions, consistent with prior research on fuzzy logic-based systems (Sinha et al., 2015; Prakash & Kumar, 2018).

Specifically, the ST-FPID controller achieved a 21°C temperature setpoint within 2 seconds without overshoot, compared to the PID controller's 6-second, 2.5°C overshoot. In humidity control, the ST-FPID controller reached 60% humidity within 2 seconds with no overshoot, outperforming the PID controller that took 6 seconds and initially overshoot to 65%. These results affirm the robustness of ST-FPID controllers in managing temperature and humidity fluctuations effectively in dynamic agricultural environments (Jones et al., 2019; Sinha et al., 2015).

Further analysis of maximum overshoot, steady-state error, and Integral Average Error (IAE) demonstrated the ST-FPID controller's consistent advantage over the PID controller under varying disturbances, ensuring precise and stable control amidst environmental fluctuations, consistent with adaptive control strategies (Smith et al., 2018; Chen et al., 2020).

Theoretical predictions and simulations confirmed the dynamic adaptability and effectiveness of ST-FPID controllers, showing improved performance metrics compared to projections. Sensitivity analysis of gain parameters and environmental factors validated the controller's robustness, suggesting its potential to optimize greenhouse conditions despite external disturbances (Sinha et al., 2015; Prakash & Kumar, 2018).

In conclusion, the ST-FPID controller's superior performance in regulating greenhouse climate conditions supports its practical implementation for enhancing crop yields and energy efficiency. This study contributes to advancing precision agriculture technologies by integrating fuzzy logic into PID controllers, emphasizing the need for further research to optimize adaptive control strategies amidst evolving environmental challenges.

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